

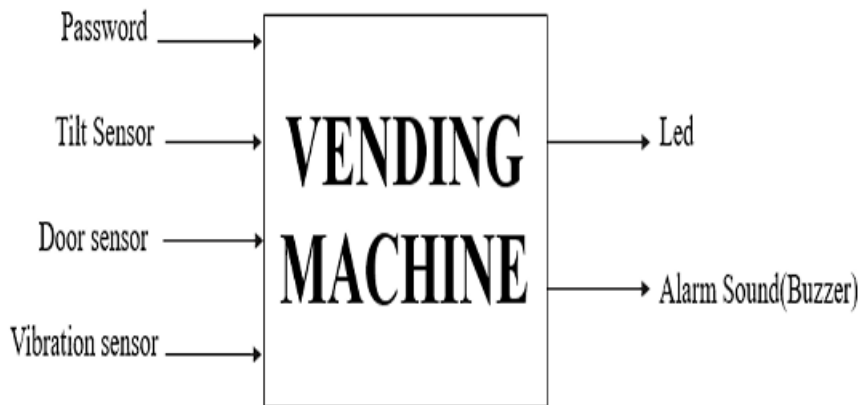
Project Overview

Vending machines are widely deployed in public locations and are often vulnerable to unauthorized access, theft, and physical tampering. This project presents a Verilog-based security system designed to improve the protection of vending machines through password authentication and sensor-based monitoring.

The system continuously monitors various security conditions and activates an alarm whenever unauthorized activity is detected.

The design was implemented using Finite State Machine (FSM) methodology and verified through simulation.

Figure 1: System Block Diagram



Description

The block diagram illustrates the overall architecture of the vending machine security system. Authentication logic, sensors, FSM controller, and alarm modules work together to monitor machine security and respond to violations.

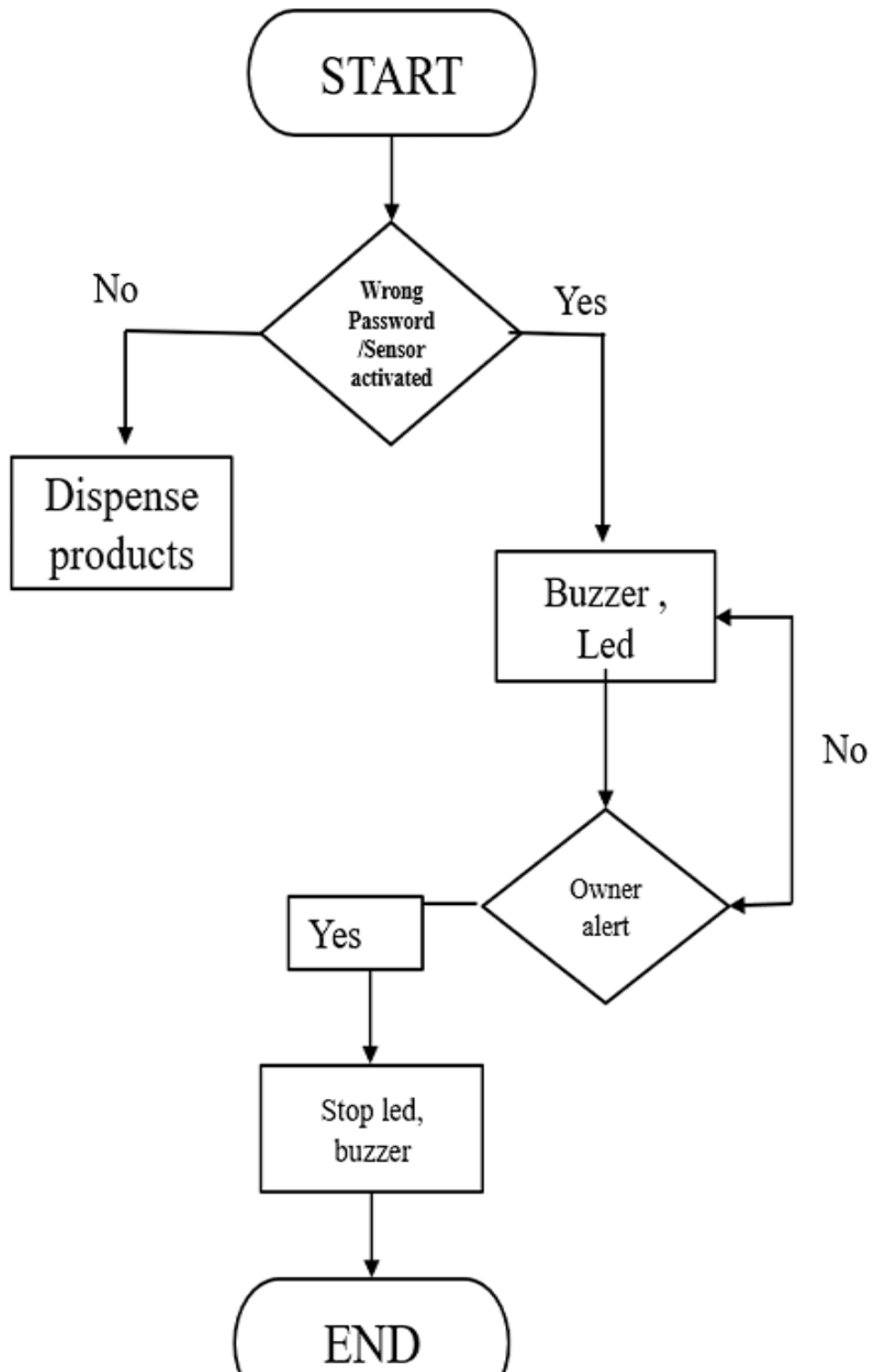
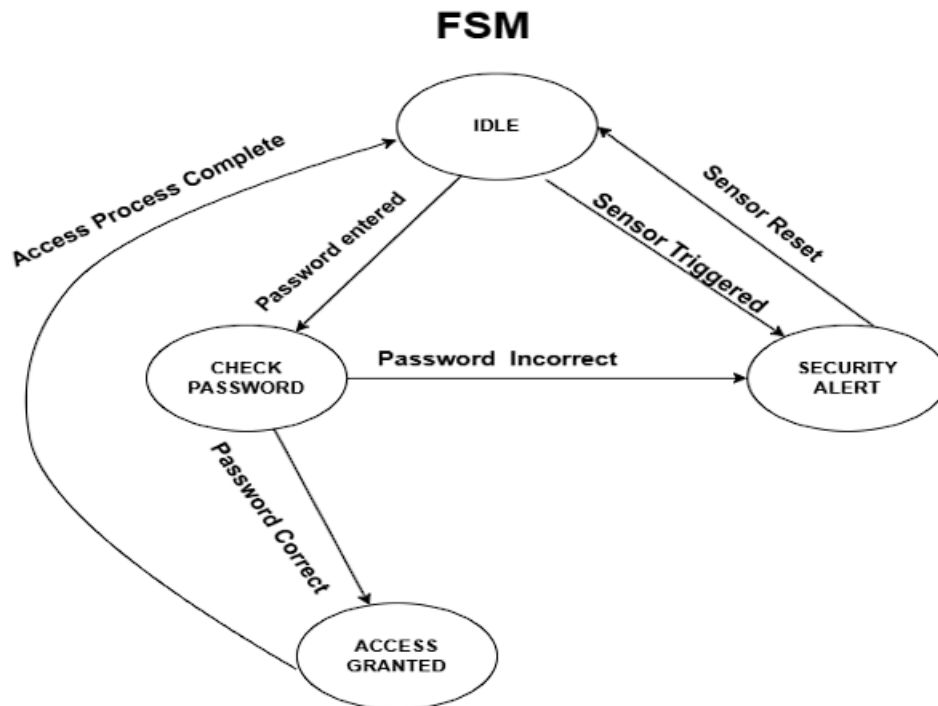


Figure 2: System Flowchart

Description

The flowchart shows the sequence of operations performed by the system. It begins with authentication and continuously monitors sensor inputs before deciding whether to remain in normal operation or activate the alarm.

Figure 3: Finite State Machine (FSM)



Description

The FSM controls system behavior by managing transitions between authentication, monitoring, access, and alarm states. State-based implementation improves reliability and simplifies verification.

The timing diagram displays the following signals and their behavior over time:

- clk**: A periodic square wave clock signal.
- reset**: A signal that starts at 1 and transitions to 0 at approximately 20,000 ns.
- buzzer**: A square wave signal that is active (1) during several intervals, including around 35,000 ns, 65,000 ns, 85,000 ns, 120,000 ns, and 155,000 ns.
- led**: A square wave signal that is active (1) during several intervals, including around 35,000 ns, 65,000 ns, 85,000 ns, 120,000 ns, and 155,000 ns.
- security_alert**: A square wave signal that is active (1) during several intervals, including around 35,000 ns, 65,000 ns, 85,000 ns, 120,000 ns, and 155,000 ns.
- access_granted**: A signal that is active (1) for a short duration around 25,000 ns.
- door_sensor**: A signal that is active (1) for a short duration around 125,000 ns.
- motion_sensor**: A signal that is active (1) for a short duration around 80,000 ns.
- vibration_sensor**: A signal that is active (1) for a short duration around 140,000 ns.
- password_input[3:0]**: A 4-bit bus signal that remains at 0 throughout the shown time interval.

The simulation waveform verifies correct system operation. Password validation, state transitions, sensor detection, and alarm activation were successfully tested during simulation.

- Correct FSM operation
- Successful password authentication
- Accurate sensor response
- Alarm activation during violations
- Stable system performance

The system successfully demonstrates a Verilog-based security architecture capable of detecting unauthorized access and responding through alarm generation. The FSM-based design provides a reliable and scalable foundation for future smart security applications.